

Matrix inequalities and norms

Sharp parameter dependence for structured Gaussian operator norms

Difficulty: challenging **Importance:** interesting to the community**Status:** Partially resolved**Topic:** Expected norms of random matrices with unequal entry variances

Rating rationale: The expectation is characterized for fixed norm exponents, but making the constant uniform requires sharper control as the exponents and dimensions vary. Such estimates quantify amplification by random matrices under different input and output norms.

Statement

Let m, n be positive integers, $1 \leq p \leq 2 \leq q \leq \infty$, and $A = (a_{ij}) \in \mathbb{R}^{m \times n}$. Let p^* satisfy $1/p + 1/p^* = 1$, with $p^* = \infty$ when $p = 1$, and set $L(k) = \max\{1, \ln k\}$ for integers $k \geq 1$. For independent standard real Gaussian variables g_{ij} , set $G_A = (a_{ij}g_{ij})$. Define

$$\|B\|_{p \rightarrow q} = \sup_{\|x\|_p \leq 1} \|Bx\|_q, \quad D_1 = \max_{1 \leq i \leq m} \|(a_{ij})_{j=1}^n\|_{p^*}, \quad D_2 = \max_{1 \leq j \leq n} \|(a_{ij})_{i=1}^m\|_q.$$

Conjecture (Latała–Strzelecka). There is an absolute constant $C > 0$, independent of m, n, p, q, A , such that

$$\mathbb{E}\|G_A\|_{p \rightarrow q} \leq C \left[\sqrt{\min\{p^*, L(n)\}} D_1 + \sqrt{\min\{q, L(m)\}} D_2 + \mathbb{E} \max_{i,j} |a_{ij}g_{ij}| \right].$$

Use the usual maximum norm for an ℓ_∞ vector norm and $\min\{\infty, L(k)\} = L(k)$. The conjecture is a uniform upper bound; it does not assert a matching lower bound with these same three terms for every variance profile.

Known cases and numerical significance

The spectral case $p = q = 2$ is known. For $a_{ij} = 1$ throughout, the two dimension-capped square-root factors have the correct order. The same paper proves comparison with $D_1 + D_2 + \mathbb{E} \max_{i,j} |a_{ij}g_{ij}|$ using constants allowed to depend on p, q . That theorem resolves the older Guédon–Hinrichs–Litvak–Prochno conjecture, while the displayed universal constant remains the new target.

The matrix G_A models independent random entry errors with a prescribed variance profile. Its induced norm measures worst-case amplification from an input ℓ_p norm to an output ℓ_q norm. The parameter dependence matters when the norms themselves vary with dimension.

References and status check

- R. Latała and M. Strzelecka, *Operator $\ell_p \rightarrow \ell_q$ norms of Gaussian matrices*, *Advances in Mathematics* 501 (2026), article 111097. DOI; current arXiv v3, dated 2026-06-09. Conjecture 5 is the displayed upper bound; the notation preceding Conjecture 1 defines the absolute implicit constant and the truncated logarithm. Theorem 2 and Remarks 3–4 distinguish the proved comparison from the conjectured parameter dependence.
- R. Latała, R. van Handel and P. Youssef, *The dimension-free structure of nonhomogeneous random matrices*, *Inventiones Mathematicae* 214 (2018), 1031–1080. The spectral-norm result is identified as reference 15 in the preceding source.

On 2026-09-11, checked the current arXiv version and published full text, the authors' institutional publication records, and exact-title, author/Conjecture-5, sharp-parameter-dependence, proof and counterexample searches through 2026. No full resolution of Conjecture 5 was located. Statements that the older Conjecture 1 has been solved do not settle this uniform-constant refinement. This is a bounded literature check.